Paper 08 - Lattice Closure Template

CQE-paper-08
CQECMPLX Formal Paper Series
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Construction Basis

Use high-dimensional lattice analogs as closure templates for transport without claiming global universality before local derivation.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- promoted formal paper: production\formal-papers\CQE-paper-08\FORMAL_PAPER.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-08\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-08\verify_lattice_closure_template.py
- receipt: production\formal-papers\CQE-paper-08\lattice_closure_template_receipt.json (Local lattice closure template: pass)
- body: production\papers\CQE-paper-08\PAPER-BODY.md
- source: production\papers\CQE-paper-08\SOURCE.md
- formal: production\papers\CQE-paper-08\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-08\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-08\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-08.md

Paper 08 - Lattice Closure Template

Abstract

Paper 08 formalizes the lattice closure template used by the CQECMPLX suite after the discrete-continuous bridge. The theorem is deliberately proof-first and local: the code chain

```
1 -> 3 -> 7 -> 8 -> 24
```

with powered terminal

```
1^2 = 1, 3^2 = 9, 7^2 = 49, 8 x 9 = 72
```

is a verified closure scaffold for transporting local states into higher dimensional lattice forms. The paper proves the local combinatorial facts that the verifiers certify: Fano/Hamming identity at dimension 7, extended Hamming self-duality and double-evenness at dimension 8, Golay ingredients and the `24 = 3 x 8` carrier at dimension 24, and the `72 = 8 x 9` powered sheet bound. It does not promote the rootless Leech landing, Gamma72 glue action, or cold-start fingerprint map as closed theorems; those are recorded as explicit audit boundaries.

Proof/Exposure Hierarchy

The proof-carrying content of this paper is the mathematics: the definitions, lemmas, constructions, examples, and receipts that establish the claimed result. Paper 00, hand routes, analog tools, workbook language, and obligation ledgers are supplemental validation and exposure layers. They exist to make the math inspectable, reproducible, and accessible without requiring a particular software stack. The hand route is not the purpose of the paper; it is a way to expose the same state transitions with ordinary marks, tokens, lines, or any equivalent physical substitute.

Definitions

A **lattice closure template** is a sequence of finite code or lattice objects that lets a local state carrier be lifted into a larger transport frame while preserving the proof boundary of every step.

The Paper 08 template is:

```
D1 raw bit                 -> 1
S3 / three-cell carrier    -> 3
Fano / Hamming / octonion  -> 7
extended Hamming / E8 seed -> 8
Golay / Leech ingredient   -> 24
Nebe sheet bound           -> 72
```

A **local certified fact** is a claim checked at a fixed dimension by a finite verifier.

A **global landing claim** is a stronger statement that a local certified fact has been glued into a terminal lattice object such as the rootless Leech lattice or a Gamma72 action. Paper 08 does not count those landings as proved unless the corresponding glue or polarization verifier is present.

A **sheet bound** is the maximum orbit distance expressible inside the current sheet before a new anchor must be introduced. The Paper 08 verifier records `K = 9`.

Main Claim

Theorem 8.1, Local Lattice Closure Template. The finite code chain `(1, 3, 7, 8, 24)` and powered terminal `72 = 8 x 9` provide a verified local closure template for CQECMPLX transport: every admitted rung is backed by a finite combinatorial check, and every unproved global landing is preserved as an explicit proof boundary rather than erased.

Proof

The verifier establishes the following local facts.

- The `(7,4,3)` Hamming code has 16 codewords, minimum weight 3, weight distribution `{0:1, 3:7, 4:7, 7:1}`, and exactly seven weight-3 codewords whose supports are the seven Fano-plane lines. Therefore the Hamming code, Fano plane, and octonion-imaginary incidence structure are identified at dimension 7.
- The `(8,4,4)` extended Hamming code has 16 codewords, minimum weight 4, weight distribution `{0:1, 4:14, 8:1}`, all weights divisible by 4, and pairwise self-orthogonality. Therefore it supplies the self-dual doubly-even binary code used as the E8 Construction A seed.
- The Golay generator layer has 12 length-24 rows, generator weights divisible by 4, zero generator-pair orthogonality errors, and the carrier geometry `24 = 3 x 8`. Therefore it supplies the binary Golay ingredients and three-copy D4 carrier geometry needed for later Leech-facing work.
- The powered chain satisfies `1^2 = 1`, `3^2 = 9`, `7^2 = 49`, and `8 x 9 = 72`. The verifier records `sheet_K_bound = 9` and the dimension 72 extremal minimum norm calculation `2 * floor(72/24) + 2 = 8`.
- The Gamma72 contract verifier checks 260 payloads and confirms exact round trips into three Leech sheets. The same verifier records `polarization_matrices_supplied = false` and `gamma72_landing_proved = false`.

Together these checks prove the local closure template and its audit boundary. The theorem does not require the global Leech or Gamma72 landing to be closed; instead, it proves that the local lattice closure surface is valid and that unproved landings remain visible. QED.

Relation to Earlier Papers

Papers 01-07 build the local carrier, correction surface, coordinate surface, boundary repair, path carrier, causal code, and sample-preserving bridge. Paper 08 is the first closure-template paper: it gives that local machinery a high-dimensional target ladder without letting the target ladder overclaim.

The result is a bridge from local proof mechanics into reusable lattice closure:

```
LCR carrier
-> correction surface
-> D4/J3 coordinate surface
-> repaired path carrier
-> causal receipt
-> discrete-continuous bridge
-> lattice closure template
```

Falsifier

The verifier rejects these overclaims:

```
"Golay ingredients alone prove the rootless Leech landing"
"three Leech-sheet round trips prove the Gamma72 polarization action"
"the chain proves that no other higher-dimensional closure template exists"
```

Those statements exceed the certified local checks. Paper 08 proves the local closure scaffold and records the stronger landings as audit boundaries.

Code Reconstruction

The production verifier is:

```
production/formal-papers/CQE-paper-08/verify_lattice_closure_template.py
```

It imports the package verifiers:

```
lattice_forge.lattice_codes.verify_lattice_code_chain
lattice_forge.lattice_codes.verify_hamming_7_fano
lattice_forge.lattice_codes.verify_extended_hamming_8
lattice_forge.lattice_codes.verify_golay_24
lattice_forge.lattice_codes.verify_powered_chain
lattice_forge.nebe_gamma72.verify_nebe_gamma72_contract
```

It verifies:

```
1. Fano/Hamming identity passes.
2. Extended Hamming/E8 seed checks pass.
3. Golay ingredient and 24 = 3 x 8 checks pass.
4. Powered 72 sheet-bound checks pass.
5. Gamma72 three-sheet transport passes while landing proof remains false.
6. Leech and Gamma72 overclaims are rejected.
```

Open Audit Boundaries

- A rootless Leech landing proof requires an explicit glue-action verifier.
- A Gamma72 landing proof requires selected Hermitian polarization matrices.
- A cold-start map from arbitrary depth to a Niemeier/Leech fingerprint

remains outside this paper.

- The claim that this is the only possible closure chain is not part of the

theorem; Paper 08 proves this chain as the suite's active closure template.

Conclusion

Paper 08 closes the first eight-paper window by giving CQECMPLX a verified lattice closure scaffold. The proof is not that every global lattice landing has been solved. The proof is sharper: local closure facts are certified, their transport role is explicit, and every stronger landing remains visible as a named audit boundary instead of being smuggled into the theorem.